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## 1 Introduction

Let  $X, Y \sim [-1, 1]$  be uniform iid rv.

Compute  $\mathbb{E}[X + Y | X + Y \geq 0]$ .

Ans:  $Z = X + Y$  is triangular from  $-2$  to  $2$ .  $W = Z | (Z \geq 0)$  is a triangle with support  $[0, 2]$  and  $p(z) = 1 - z/2$ . We compute

$$\begin{aligned} E[W] &= \int_0^2 (1 - w/2)w dw = \left( \frac{1}{2}w^2 - \frac{1}{6}w^3 \right) \Big|_0^2 \\ &= 2 - \frac{8}{6} = \frac{2}{3}. \end{aligned}$$

Compute  $\sigma_W$ .

Ans:

$$\begin{aligned} E[W^2] &= \int_0^2 (1 - w/2)w^2 dw = \int_0^2 \left( w^2 - \frac{1}{2}w^3 \right) dw \\ &= \left( \frac{1}{3}w^3 - \frac{1}{8}w^4 \right) \Big|_0^2 = \frac{8}{3} - 2 = \frac{2}{3} \end{aligned}$$

so  $\text{Var}(W) = E[W^2] - E[W]^2 = \frac{2}{3} - \frac{4}{9} = \frac{2}{9}$ , so  $\sigma_W = \frac{\sqrt{2}}{3} \approx 0.47$ .

Assume there are  $N$  traders. Let the initial spread  $\Delta$  be centered around 0 and assume  $\Delta$  is sufficiently small. Suppose the traders bid logically, so that  $N = N^+ + N^-$ , where the  $N^+$  traders have nonnegative, positive hands, and bought and sold. The resulting expectancy is  $\frac{2}{3}(N^+ - N^-)$  with variance  $\frac{2}{9}N$ , or STD  $\frac{1}{3}\sqrt{2N}$ . Denote this distribution by  $T$ .

Assume  $N$  is sufficiently large, so  $T$  approximately normal. The expected distance from the mean (0) for the standard normal distribution is  $\sqrt{2/\pi}$ . Thus, the expected earnings for bidding  $\frac{2}{3}(N^+ - N^-)$  is

$$P = N\Delta - \sqrt{\frac{2}{\pi}} \cdot \frac{1}{3}\sqrt{2N}$$

For  $P \geq 0$ , we wish for  $\Delta \geq \sqrt{\frac{4}{9\pi N}}$ .

Scaling to 13, we have the following table for  $\Delta$  (for  $P \geq 0$ ):

| $N$ | $\Delta$ |
|-----|----------|
| 3   | 2.8      |
| 4   | 2.4      |
| 5   | 2.2      |
| 6   | 2.0      |

Problem: There are 12 beans, one weights slightly heavier or lighter than the others. What is the minimum number of times a balance scale must be used to guarantee the determination of the abnormal bean?

Ans: Label the beans  $1, 2, \dots, 12$ .  $\{1, 2, 3\} \neq \{4, 5, 6\}$ . If  $\{1, 2, 3\} = \{4, 5, 6\}$ ,  $\{1, 2, 3\} \neq \{7, 8, 9\}$ .  
If  $\{1, 2, 3\} = \{7, 8, 9\}$